

# Independent Axial Seamount Thermomechanical Model: Progress Report

Axial Modeling Project

October 2026

## Abstract

This report records an independent PyLith implementation of the written thermomechanical method for Axial Seamount, with observational checking against authorized Ocean Observatories Initiative (OOI) records and original raw BPR channels from earlier Axial deployments. It uses no paper-produced observations, corrections, numerical results, model output, or figure data. The project verifies elastic and one-branch Maxwell responses, steady three-dimensional heat conduction, temperature-property transfer, and postprocessed failure connectivity, and synthetic three-branch historical forward checks. Central and Eastern OOI histories provide 3,927 common finite daily records from 2014 through 2026; the aggregate quality code is retained as NOT EVALUATED. Original raw NCEI BPR channels add deployments from 1987 through 2002, including the 1998 event, and raw MGDS channels add the 2002–04 Center record and further records through 2022, including the 2011 event and post-eruption overlaps. The 2002–04 South holdout has 0.658 m RMSE over 317 paired days. The first one-branch OOI check applies a static-elastic pressure fit to Maxwell mechanics and has a large Central displacement error. A separate Maxwell-kernel inversion reduces Central RMSE to 0.005 m but leaves 0.225 m Eastern holdout error. Raw 1998 and 2011 kernel checks leave South RMSE values of 0.535 and 0.713 m. The historical three-branch event checks use synthetic branch parameters and have held-out South biases of 0.329 m and 0.371 m. Additional deployment checks span 1995–2022; the 2011–13 South comparison has 1.242 m RMSE and  $-1.214$  m bias. The 2013–15 South 2 and South 1 checks have 0.358 m and 1.096 m RMSE, and the 2015–17 South 2 check has 0.265 m RMSE. The 2018–20 South 2 and 2020–22 miniBPR South 1 checks have 0.105 m and 0.043 m RMSE across 741 and 648 paired days. Sixteen further raw MGDS station holdouts span these three later windows; their RMSE values range from 0.031 m to 0.705 m. Documented noise and sediment instability exclude two 2020–22 records, while the 2018–20 West BPR lacks a drift estimate and has 0.346 m RMSE. These added spatial checks retain raw ocean and instrument effects and do not calibrate the synthetic branches or static compliance. The four-case raw BPR fits give 0.093 m Center RMSE in 1998 and 0.124 m in 2011; held-out South RMSE is 0.519–0.531 and 0.704–0.717 m. Fitted pressure minima reach  $-95$  and  $-72$  MPa. Both Center fits include observed eruption deflation, so their connected-path histories cannot independently predict eruption timing. These outputs remain provisional: the ellipsoid compliance is not mesh-converged, generalized Maxwell branch parameters are unavailable, Eq. 16 is inconsistent with its stated modulus limits, and the paper’s written thermal equation specifies zero heat production without mechanical feedback. The current work therefore supports a partial implementation record, not a reproduction of the published eruption forecasts.

## 1 Scope and data provenance

The project reconstructs model inputs from written descriptions of the study and its supplement, then implements independent solvers and checks. The main model article and publisher-served supplement are specification sources [1, 2]. No observation series, corrections, numerical results, model output,

source code, plotting scripts, or figure data produced for the paper are used. The report does not digitize plotted curves or use published numerical results. The supplement copy carries a confidential manuscript footer, so extracted parameters remain version-qualified.

OOI daily BPR depth records cover Central and Eastern Caldera. The products report relative uplift after detiding and drift correction; their aggregate quality flag is NOT EVALUATED [4]. The analysis retains flag code 2 without filtering. Common finite observations span 2014-09-05 through 2026-09-30. Raw NCEI records provide uncorrected pressure from ten Axial deployments in 1987–96 and further deployments in 1997–99 and 2000–02. WC09–WC32 use 56.25-second samples; WC51 onward uses 15-second samples. Selected MGDS files provide original `Depth`, `RawDep`, and `RawDepth(m)` channels from Center and South deployments between 2003 and 2022 [5, 6]. The 2020–22 miniBPR records are sampled every 100 seconds. The workflow excludes derived detided, filtered, and pressure-drift-corrected fields; source files and processed daily series remain outside version control. The MGDS archive uses the CC BY-NC-SA 3.0 license, and its data citation and license are preserved in the repository data notes.

## 2 Written model and implementation

The specified reservoir is an ellipsoid measuring 6 by 3 by 1 km, centered 1.6 km below the seafloor. The written thermal method is steady conduction,

$$\nabla \cdot (k \nabla T) = -Q, \quad Q = 0, \quad (1)$$

with a 30 °C/km background geotherm, a 0 °C surface, and a 1200 °C reservoir boundary. The hydrothermal conductivity law is implemented as a separate case. Because lateral and basal thermal conditions are not specified, the current finite-element calculation extends the background geotherm to all outer faces and records this as an assumption.

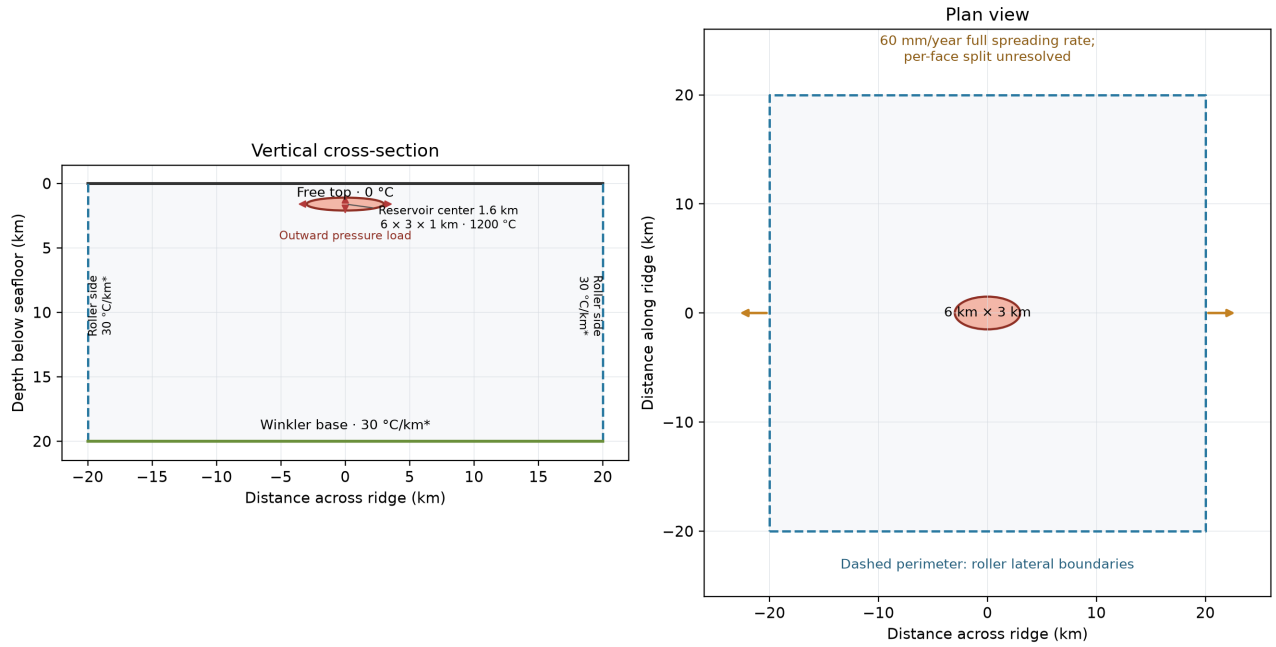
The Arrhenius viscosity law is evaluated from the temperature field. The printed temperature-dependent Young’s modulus equation increases from about 25 GPa toward 75 GPa as temperature rises, contrary to the written brittle and ductile descriptions. The implementation preserves this equation only as a diagnostic; it does not claim the trend is physically resolved. The written Maxwell model requires a generalized branch spectrum that is not fully specified. Current PyLith runs therefore use one Maxwell branch with explicitly assumed properties. The PyLith 5.0.2 material interface supplies properties and initial state through an auxiliary spatial database; a same-mesh restart can transfer state when properties change [3].

The hydrothermal thermal solve now produces an exploratory midplane map of temperature-dependent properties (Fig. 2). This slice uses the written Arrhenius viscosity and conductivity relations and evaluates Eq. 16 exactly as printed. The finite-thickness slice, outer-face geotherm, and incomplete generalized Maxwell spectrum limit comparison with the complete four-rheology panels.

The hydrothermal Picard solve converges numerically on three independently generated meshes with 2,761, 2,941, and 3,060 tetrahedra. Spatial changes do not settle at 105 common host-rock probes: the 1,200-to-1,150 m pair has a 13.45 °C probe RMSE and a 64.61 °C maximum change, while the 1,150-to-1,100 m pair has a 35.55 °C RMSE and a 324.29 °C maximum near the reservoir edge (Fig. 3). The mesh sequence is unstructured and non-nested, so these finite-probe comparisons do not establish a volume norm or formal convergence. They show that the current thermal field remains spatially sensitive near the reservoir.

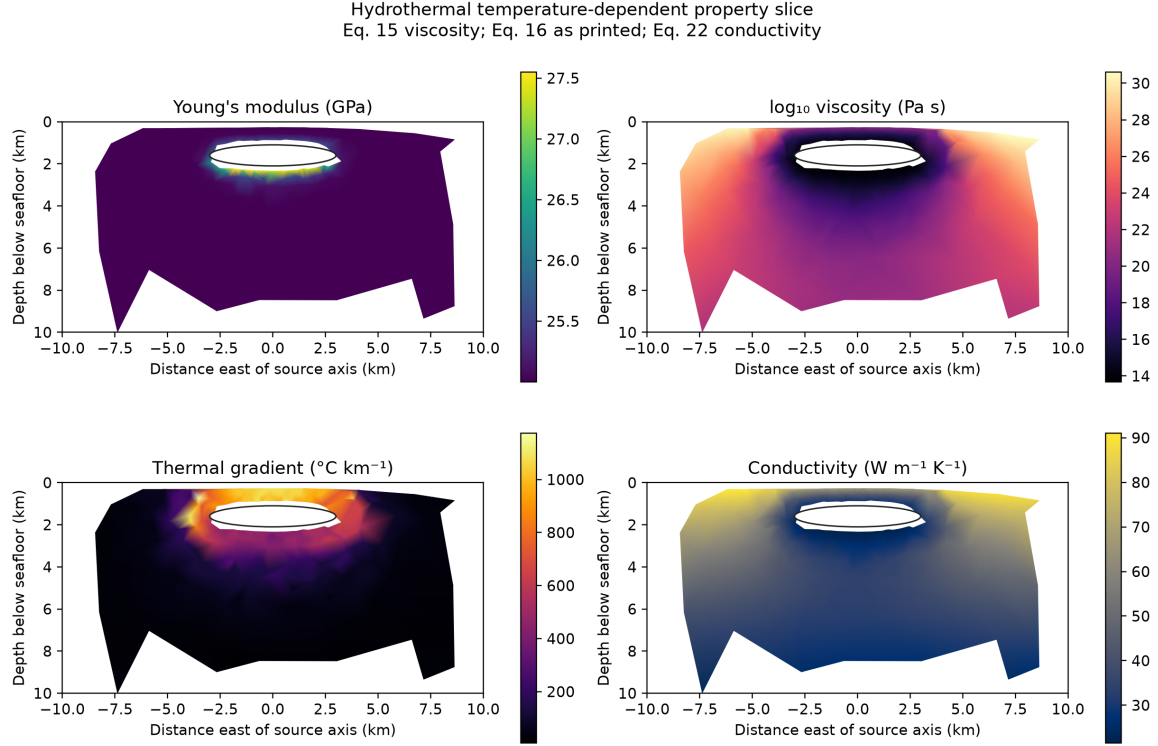
The project has verified a three-dimensional steady heat solve, a synthetic thermal-to-property-to-mechanics transfer, and same-mesh Maxwell restart. An unchanged-property segmented run differs

Axial thermomechanical model setup — schematic



The 40 × 40 × 20 km box is a project fallback; the written model extent is unavailable. \*The side and basal 30 °C/km geotherm is an explicit thermal assumption. Reservoir depth is 1.6 km to center; Winkler stiffness and the tectonic face-rate split are unresolved.

Figure 1: Schematic of the written reservoir geometry and boundary conditions. The 40 by 40 by 20 km model box is a project fallback; the ellipsoid dimensions and center depth follow the written specification. Roller sides, a Winkler base, a free top, and the 60 mm/year full spreading rate are labeled without assigning an unresolved per-face velocity. The 30 °C/km side and basal thermal conditions are project assumptions. Generated by `make model-setup-schematic`; no published figure or geological data were used.



Cell-center projection within  $|y| \leq 0.7$  km; white ellipse marks the reservoir. Side and basal temperatures use the assumed  $30^{\circ}\text{C/km}$  geotherm. Eq. 16 conflicts with the stated brittle and ductile limits.

Figure 2: Hydrothermal temperature-dependent properties on a finite-thickness slice around the model  $y=0$  plane. Cell-center values show Young's modulus from Eq. 16 as printed, Arrhenius viscosity, the finite-element temperature gradient, and Eq. 22 conductivity. The white ellipse marks the reservoir. Side and basal temperatures extend the assumed  $30^{\circ}\text{C/km}$  geotherm. Generated by `make thermal-property-slices`; Eq. 16 remains an unresolved transcription diagnostic.

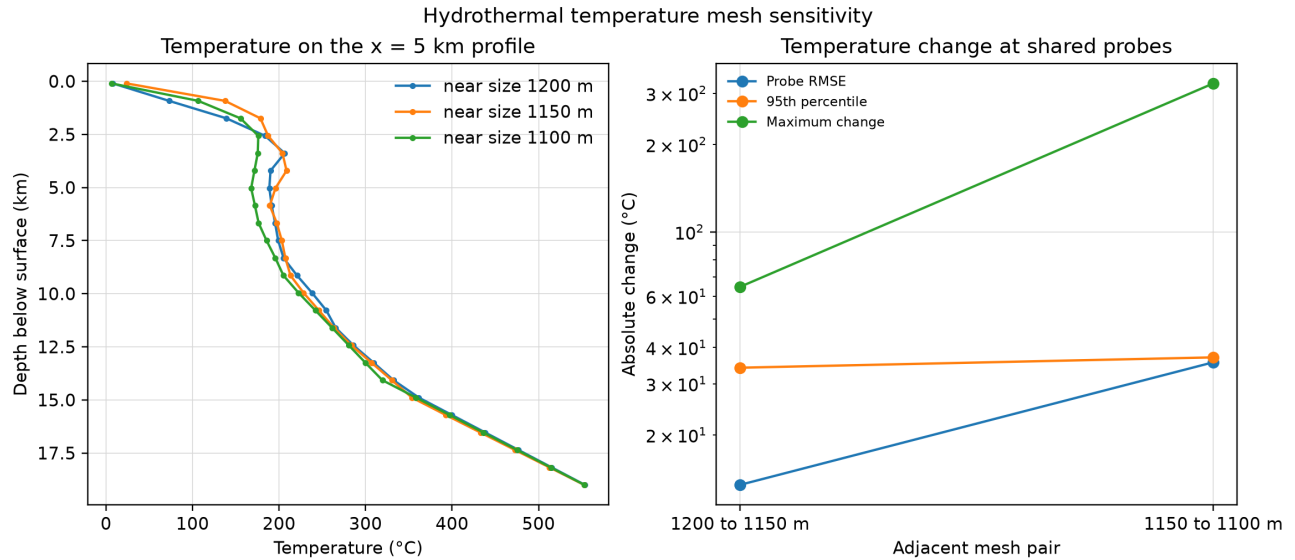


Figure 3: Hydrothermal temperature sensitivity across three bounded tetrahedral meshes. The left panel shows interpolated temperatures along a common vertical profile. The right panel shows RMSE, 95th-percentile, and maximum absolute changes across 105 common host-rock probes for adjacent meshes. Meshes are independently generated and non-nested; the probe changes diagnose spatial sensitivity but do not establish convergence. Generated by `make thermal-mesh-sensitivity`.

from a continuous run by at most  $1.612 \times 10^{-8}$  in displacement, stress, total strain, and viscous strain. A separate uniform 1200 °C material swap carries viscous strain across the restart boundary and changes the final displacement by 50.08 percent relative to the uniform-property run. This verifies the restart interface; the artificial uniform-temperature swap is not a coupled physical case.

A manufactured check transfers an affine temperature field from a six-tetrahedron source mesh to the 2,761-tetrahedron mechanics mesh, with a maximum interpolation error of  $1.14 \times 10^{-13}$  °C. The smoke case also solves the written hydrothermal model on a distinct 3,060-tetrahedron ellipsoid mesh. Sampling at mechanics-cell centers yields finite temperatures from 8.769 to 1,066.240 °C. PyLith completes a two-second Maxwell solve using the mapped heterogeneous material database and writes finite stress and viscous strain; peak stress is  $1.75811 \times 10^7$  Pa. The smoke case verifies point sampling of a physical steady field and material-database use across meshes. It does not test conservative transfer, time-varying properties, or two-way thermal-mechanical feedback.

The paper’s thermal method specifies  $Q = 0$  and gives no stress- or deformation-generated heat term. The roadmap’s mechanics-to-thermal return coupling therefore lacks a written physical law. The current one-way OOI diagnostic does not add an invented heat source. This prevents a complete two-way reproduction until the return coupling is specified or approved as a separate exploratory extension.

### 3 Independent observations and model checks

Figure 4 shows the available Central and Eastern OOI uplift records. Each instrument uses its own first-sample reference. The plot is an observation-only partial record: it has no raw pre-2014 values, eruption markers, or earthquake counts.

The original raw NCEI channels add intermittent Axial deployment records from 1987 through 2002, while MGDS raw channels extend coverage from July 2002 through 2022. For the NeMO 2002–04 Center

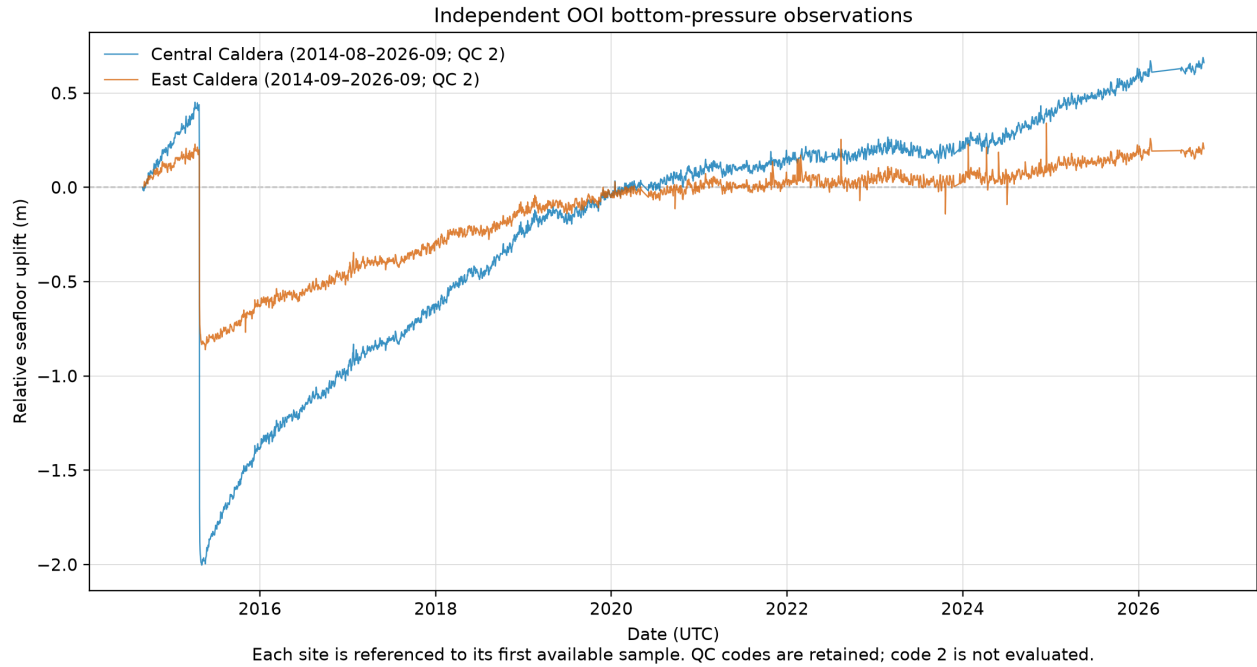


Figure 4: Daily OOI relative uplift at Central and Eastern Caldera from each site’s first available record. Generated by `make bpr-observation-plot`; aggregate quality code 2 (NOT EVALUATED) is retained. The data show the observed interval available for independent model checking.

file, MGDS identifies its `DriftCorrRawDep` field as unchanged because the recorded drift correction was zero. The analysis excludes its tide-subtracted and filtered fields. The deployment-context figure places every deployment on its own first-sample baseline; gaps between deployments are not interpolated, and no tide, ocean, or sensor pressure-drift correction is applied. WC09–WC32 were sampled every 56.25 seconds; later NCEI and moored MGDS records use 15-second samples, while the 2020–22 miniBPRs use 100-second samples. The unstable 2017–18 Center record is displayed for context and excluded from model forcing.

An independent Fox archive from MGDS (IEDA/322344; doi:10.1594/IEDA/322344) for the same 1997–98 Center and South instruments agrees with NCEI relative uplift over 309 Center and 365 South days, with correlations above 0.999999 and RMSE values of 0.050 and 0.016 m. This checks raw-channel handling across archives; it does not add independent station coverage. The archive’s detided and low-pass-filtered fields are excluded.

An elastic Mogi point source is calibrated to Central uplift and checked against the Eastern record. The inferred pressure spans  $-2.96$  to  $+1.06$  GPa; Eastern RMSE is 0.0594 m with 0.995 correlation. The spatial correlation does not validate this pressure scale. It shows that the point-source proxy cannot supply the intended historical ellipsoid pressure model.

The raw historical channels add Center-to-South checks at the 1998 and 2011 eruptions, a 1995–96 pre-eruption overlap, and three later deployment overlaps. The 1995–96 WC68/WC69 overlap spans 338 paired daily values. Its held-out South prediction has 0.180 m RMSE,  $+0.159$  m bias, and 0.876 correlation; the raw record’s slow South trend is not captured by the static model. The later pairs span 614 paired days in 2003–05, 572 in 2007–09, and 731 in 2011–13. Static Mogi predictions have South RMSE values of 0.134, 0.156, and 0.387 m, with respective biases of  $+0.113$ ,  $+0.136$ , and  $+0.363$  m. These channels retain tides, oceanographic variability, and sensor drift, and the fits omit viscoelastic memory. They extend observation checking across the pre-OOI interval; they are not calibrated pressure

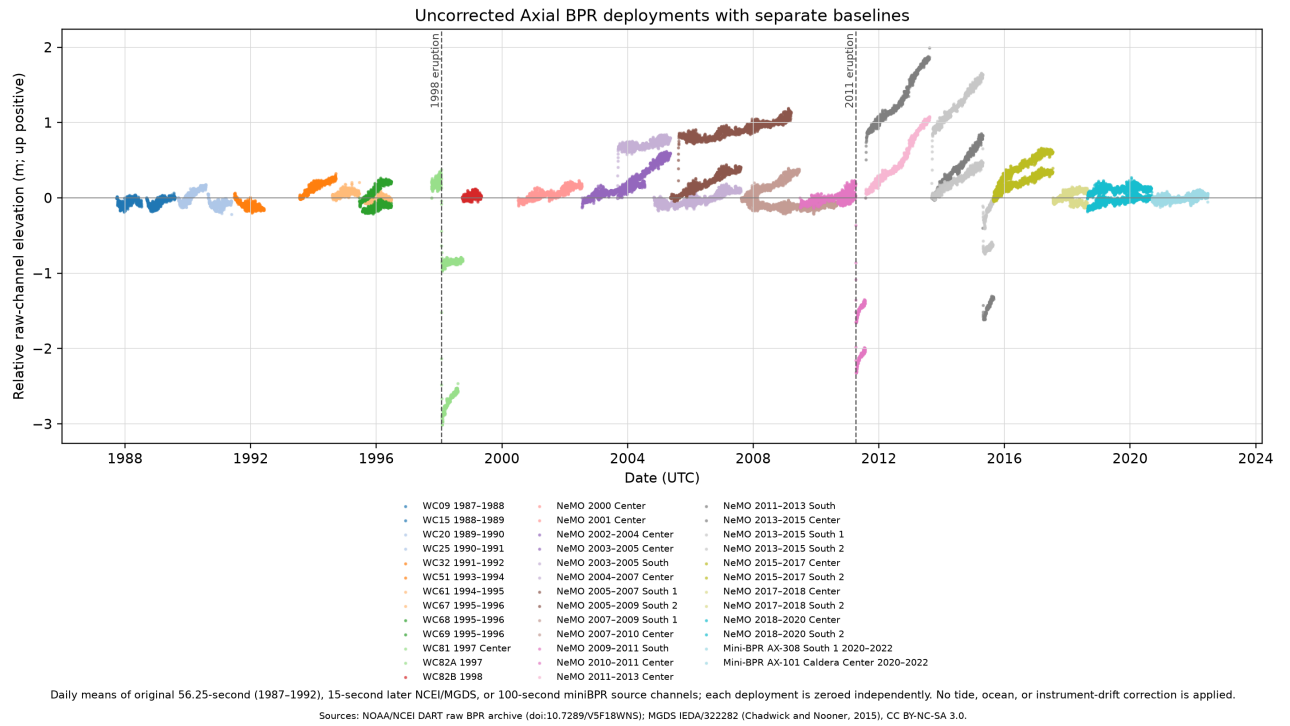


Figure 5: Separate-baseline daily means of original raw non-OOI BPR channels from 1987 through 2022. The plot shows intermittent deployment coverage; each segment is zeroed independently and remains uncorrected for tides, ocean variability, and instrument drift. Generated by `make bpr-historical-check`. Sources: NOAA/NCEI DART raw BPR archive (doi:10.7289/V5F18WNS) and MGDS IEDA/322282 (Chadwick and Nooner, 2015). MGDS-derived values are distributed under CC BY-NC-SA 3.0.

histories or eruption forecasts.

The three-branch historical forward checks use first-common-day zeroing and infer each daily pressure from the static Center compliance. In the 1998 pair, the Center and held-out South RMSE values are 0.186 m and 0.503 m; the 2011 values are 0.114 m and 0.680 m. South biases are +0.329 m and +0.371 m, respectively. From 2013–15, Center forcing predicts South 2 with 0.358 m RMSE; South 1, held out independently, has 1.096 m RMSE and  $-0.988$  m bias. The 2015–17 South 2 holdout has 0.265 m RMSE and  $-0.245$  m bias. The inferred pressure ranges reach  $-50.7$  to  $+26.7$  MPa and 0 to  $+20.8$  MPa. These checks use synthetic branch reference viscosities and fractions, retain raw ocean and instrument effects, and inherit the nonconverged static compliance. They test the historical loading path without calibrating rheology or producing a hindcast.

The saved stress histories for all twelve raw-BPR-driven windows are also evaluated with the current Mohr–Coulomb proxy: cohesion is 1 MPa, friction angle is 25 degrees used directly as  $\phi$ , and pore pressure is zero. A tensile cutoff is not applied to the shear path. The resulting cavity-to-surface paths appear within the first 196 days of all twelve windows, including the ten intervals without an eruption. In 2002–04, path onset is bracketed at 53.64 days; its first saved path appears at day 56. In 1998 a path is already present at the first saved output, so onset is only bounded above by seven days. In 2011 linear stress interpolation brackets onset at 17.61 days after the first common BPR day. The 2013–15 and 2015–17 paths appear at 34.75 days and by the first saved day, respectively; the 2018–20 and 2020–22 paths first appear at 21 and 35 days. The same proxy therefore fails to distinguish the eruption windows from inter-eruption intervals and is not calibrated to eruption timing. The written criterion also requires tensile failure at the reservoir, but tensile strength is unspecified and is not assigned. At saved records with a connected path, maximum cavity tension is 94.0 MPa in 1998 and 71.1 MPa in 2011, compared with 58.3 MPa across the ten inter-eruption windows. Using unrounded output values, strengths in  $(58.3221, 71.0951]$  MPa would separate these windows only for this saved-record diagnostic and its synthetic branch properties, zero pore pressure, direct friction interpretation, raw uncorrected channels, and unconverged mesh. This conditional interval is not a measured or calibrated rock strength. The joint condition is checked only at saved PyLith records.

| BPR window | First path time | Saved records with a path | Max. tension on path (MPa) |
|------------|-----------------|---------------------------|----------------------------|
| 1995–96    | $\leq 7$ days   | 46/49                     | 6.08                       |
| 2002–04    | 53.64 days      | 30/46                     | 6.94                       |
| 1998       | $\leq 7$ days   | 44/44                     | 94.00                      |
| 2003–05    | 25.38 days      | 83/88                     | 32.30                      |
| 2005–07    | 195.84 days     | 71/117                    | 14.18                      |
| 2007–09    | 67.86 days      | 68/83                     | 10.96                      |
| 2011       | 17.61 days      | 30/47                     | 71.10                      |
| 2011–13    | $\leq 7$ days   | 105/106                   | 58.32                      |
| 2013–15    | 34.75 days      | 96/102                    | 50.50                      |
| 2015–17    | $\leq 7$ days   | 97/98                     | 36.00                      |
| 2018–20    | 21 days         | 98/106                    | 13.69                      |
| 2020–22    | 35 days         | 16/93                     | 6.11                       |

Table 1: Provisional joint failure diagnostics on the twelve historical three-branch stress histories. Time is measured from each independently zeroed BPR window. Thresholds use 1 MPa cohesion, 25-degree friction angle, and zero pore pressure. Maximum tension is the maximum cavity stress on saved records that also have a connected path; no tensile strength is assigned. Onsets after the first saved record are interpolated linearly between saved PyLith stress records; paths already present in the first output are bounded by seven days.

A bounded time-step refinement shows that these first-path estimates depend on the integration interval.

In the 2011 history, the interpolated first crossing moves from 17.61 days at seven-day steps to 2.29 days at one day and 2.28 days at half- and quarter-day steps. The 2002–04 quiet-period estimate shifts from 53.64 days at seven-day steps to 26.94 days at one day and 26.93 days at half- and quarter-day steps. The full 80-day quarter-day cases completed within the five-minute per-run limit. At finer steps, both paths appear and disappear repeatedly. These results show that coarse sampling misses short-lived paths, while the refined first crossing remains a diagnostic under synthetic branches, static pressure compliance, and the nonconverged mesh, not a sustained eruption time.

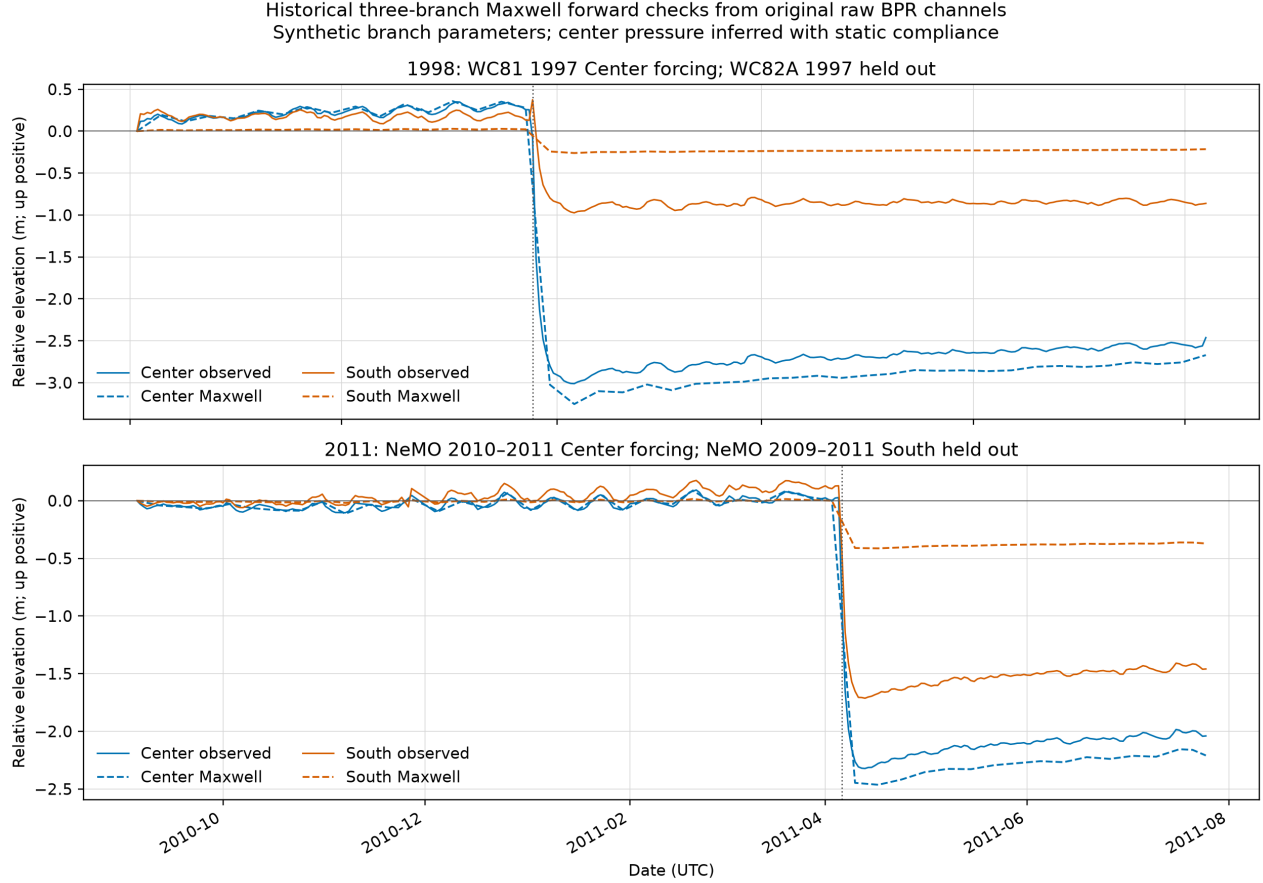


Figure 6: Three-branch generalized Maxwell forward checks against raw daily BPR observations for the 1998 and 2011 Center/South deployment pairs. Center pressure is inferred from static elastic compliance; South is held out. Synthetic branch properties, first-common-day zeroing, uncorrected raw records, and nonconverged compliance limit interpretation. Generated by `make historical-generalized-maxwell-check`.

The 2011 raw-BPR check also carries one three-branch Maxwell stress history from the September 2010 event overlap through the August 2013 replacement deployment. The Center records do not overlap and have a five-day gap, so the run holds inferred pressure at its terminal pre-gap value and baselines each instrument segment independently. The event South holdout has 0.680 m RMSE, +0.371 m bias, and 0.997 correlation. The 2011–13 South holdout has 1.246 m RMSE, −1.217 m bias, and 0.991 correlation. This shared slow trend does not remove the large holdout bias; the result remains provisional under synthetic branches, raw uncorrected channels, and nonconverged compliance.

The 1998 check carries one three-branch Maxwell stress history from the October 1997 WC81/WC82 overlap through May 1999. The original WC81 Center channel ends on 7 August 1998; the run then

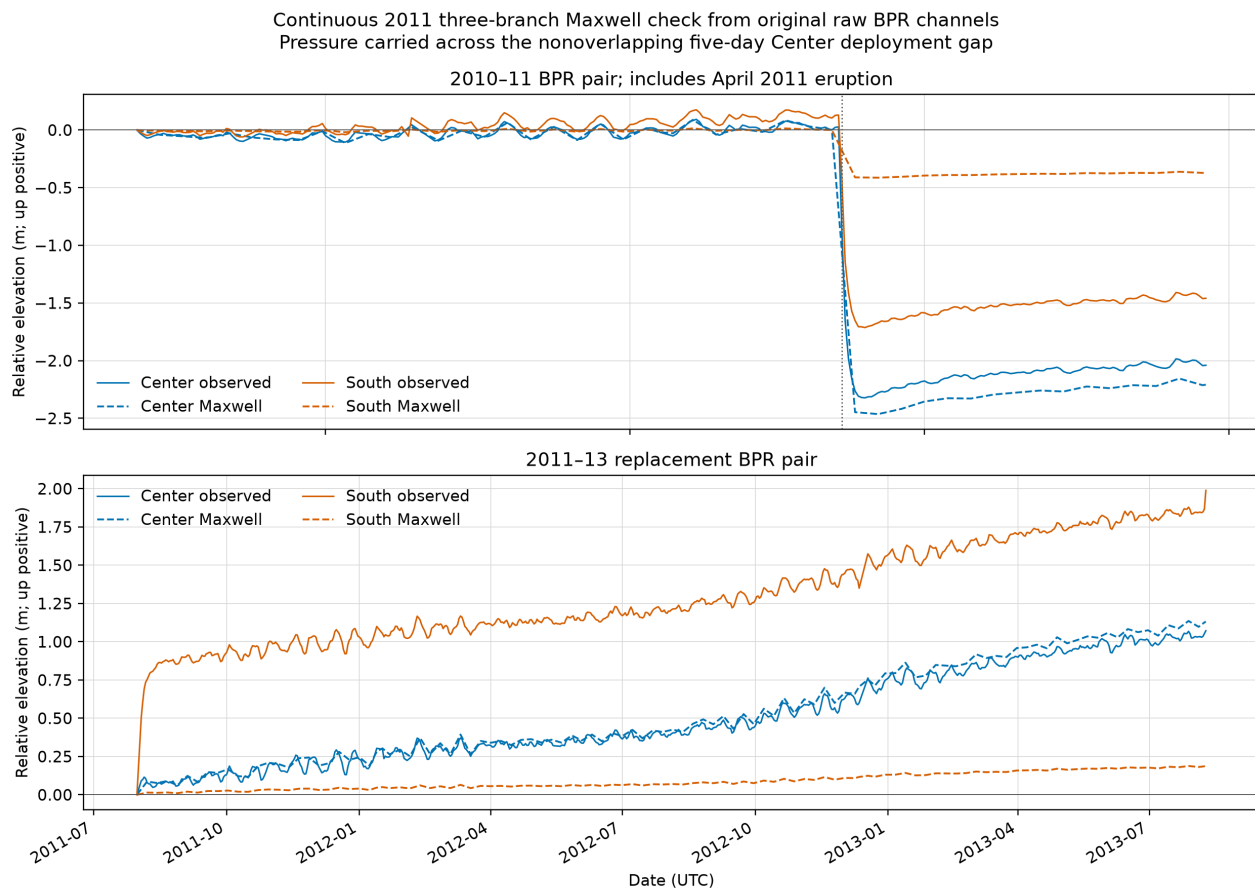


Figure 7: Continuous 2011 three-branch Maxwell check from original raw MGDS BPR channels. Each panel shows Center-forced predictions and its paired South holdout, independently baselined by deployment. The Center pressure series are joined across a five-day gap by holding the final pressure constant. The eruption date is marked in the upper panel. Synthetic branch parameters, raw instrument effects, and nonconverged compliance limit interpretation. Source channels: MGDS IEDA/322282 (Chadwick et al., 2023), CC BY-NC-SA 3.0. Generated by `make historical-generalized-maxwell-2011-continuous-check`.

holds terminal inferred pressure constant for 270 days while the WC82 South station continues. The 309-day event South holdout has 0.503 m RMSE, +0.329 m bias, and 0.995 correlation. After the Center record ends, the independently baselined South holdout has 0.063 m RMSE and -0.369 correlation. Its low RMSE reflects the small amplitude of the nearly flat model and raw follow-up record, not evidence of predictive skill. The continued pressure is an explicit assumption, and the result remains provisional under synthetic branches and nonconverged compliance.

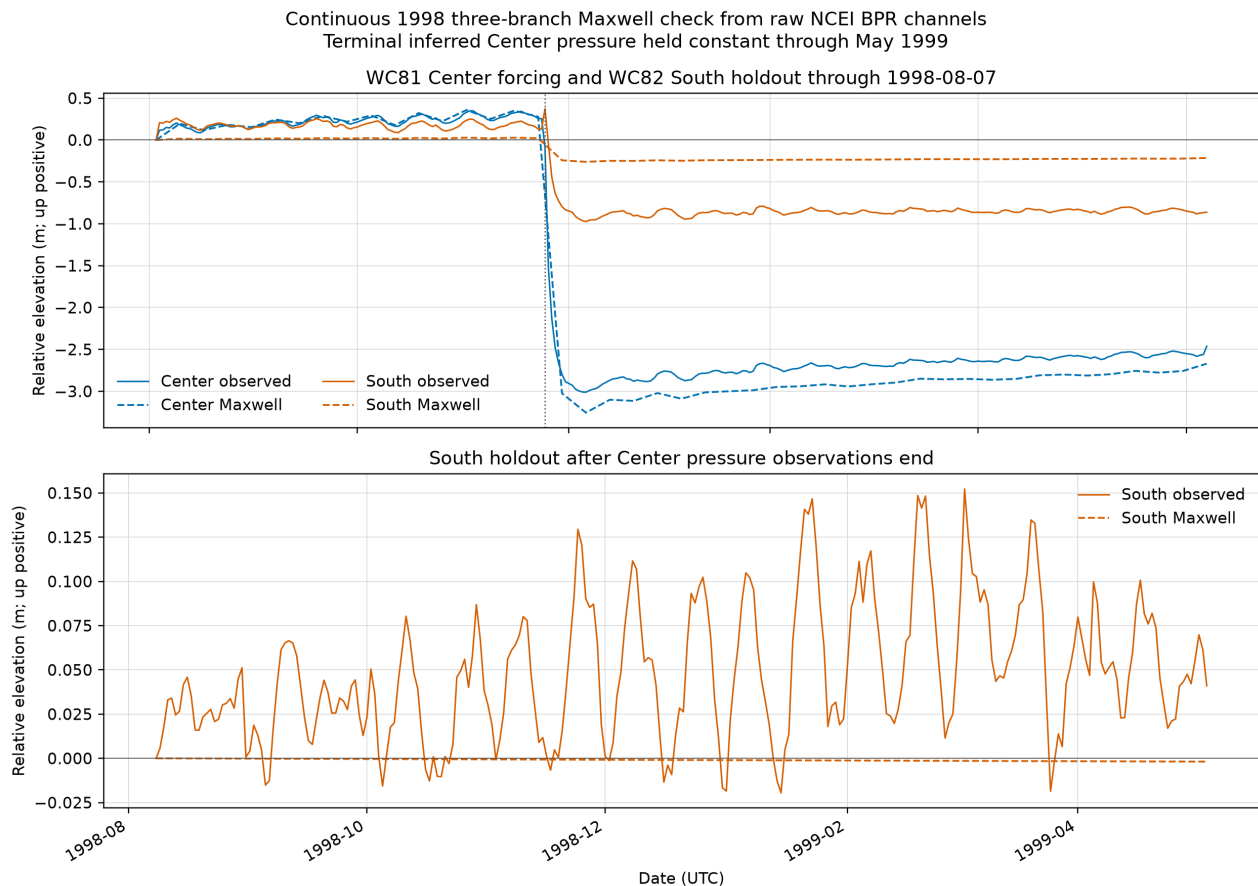


Figure 8: Continuous 1998 three-branch Maxwell check using only original NCEI WC81 Center and WC82 South absolute-pressure channels. The upper panel compares Center forcing and the event South holdout through 7 August 1998; the lower panel checks the South station through 4 May 1999 after Center pressure observations end. The two WC82 South files are aligned from their eight-day raw overlap. Terminal inferred Center pressure is held constant for 270 days. Synthetic branch parameters, raw ocean variability, and nonconverged compliance limit interpretation. Source: NCEI DART raw BPR archive, doi:10.7289/V5F18WNS. Generated by `make historical-generalized-maxwell-1998-continuous-check`.

Ten additional raw Center/South overlaps extend the three-branch check across 1995–2022. Table 2 summarizes the held-out South comparisons. Each window has its own first-day zero, and deployment gaps are left unfilled. The 2011–13 correlation of 0.991 coexists with a 1.242 m RMSE and  $-1.214$  m bias. Large residuals, changing correlations, synthetic branch parameters, and uncorrected records prevent these plots from establishing a historical calibration.

The same model runs also predict three additional raw BPR locations that were not used to define their primary Center/South pairs. WC67 is held out from the 1995–96 WC68-driven run over 338 days;

| Interval | RMSE (m) | Bias (m) | Correlation |
|----------|----------|----------|-------------|
| 1995–96  | 0.183    | −0.161   | 0.793       |
| 2002–04  | 0.658    | −0.655   | 0.366       |
| 2003–05  | 0.651    | −0.649   | 0.660       |
| 2005–07  | 0.157    | −0.135   | 0.958       |
| 2007–09  | 0.123    | −0.101   | −0.355      |
| 2011–13  | 1.242    | −1.214   | 0.991       |
| 2013–15  | 0.358    | −0.089   | 0.991       |
| 2015–17  | 0.265    | −0.245   | 0.973       |
| 2018–20  | 0.105    | −0.096   | 0.787       |
| 2020–22  | 0.043    | −0.017   | 0.788       |

Table 2: Held-out South residuals for ten raw deployment overlaps.

NeMO South 1 is held out from the 2007–09 Center-driven run over the 572 days shared with the primary window. The 2013–15 South 1 record adds a 711-day holdout. The WC67 residual is small; the later South holdouts have larger errors, including the 2013–15 South 1 bias of  $-0.988$  m. These differences show that spatial transfer varies across deployments and that every raw channel remains a separate model check.

| Held-out raw station | Window  | RMSE (m) | Bias (m) | Correlation |
|----------------------|---------|----------|----------|-------------|
| WC67                 | 1995–96 | 0.036    | −0.021   | 0.685       |
| NeMO South 1         | 2007–09 | 0.221    | −0.191   | −0.506      |
| NeMO South 1         | 2013–15 | 1.096    | −0.988   | 0.985       |

Table 3: Additional independent BPR station checks using the existing Center-driven Maxwell solutions. Neither station contributes to the pressure history used in its model run.

Sixteen further raw MGDS stations add holdouts to the 2015–17, 2018–20, and 2020–22 Center-driven solutions. Each station and its Center comparison are zeroed on their first shared day. The model uses only original **RawDep**, **RawDepth(m)**, or **Depth** channels; no data product associated with Cabaniss et al. enters the comparison. Raw ocean variability and drift remain. AX-105 in 2018–19 has no linear drift correction in the source file; the 2018–20 West record has no MPR-based drift estimate. Unknown drift remains in the 2020–22 East and North records. The 2018–20 North record is omitted because its archived coordinates conflict with its source location note. AX-303 in 2020–22 is excluded from metrics for its documented high-noise interval, and BPR West is excluded for documented sediment-site instability.

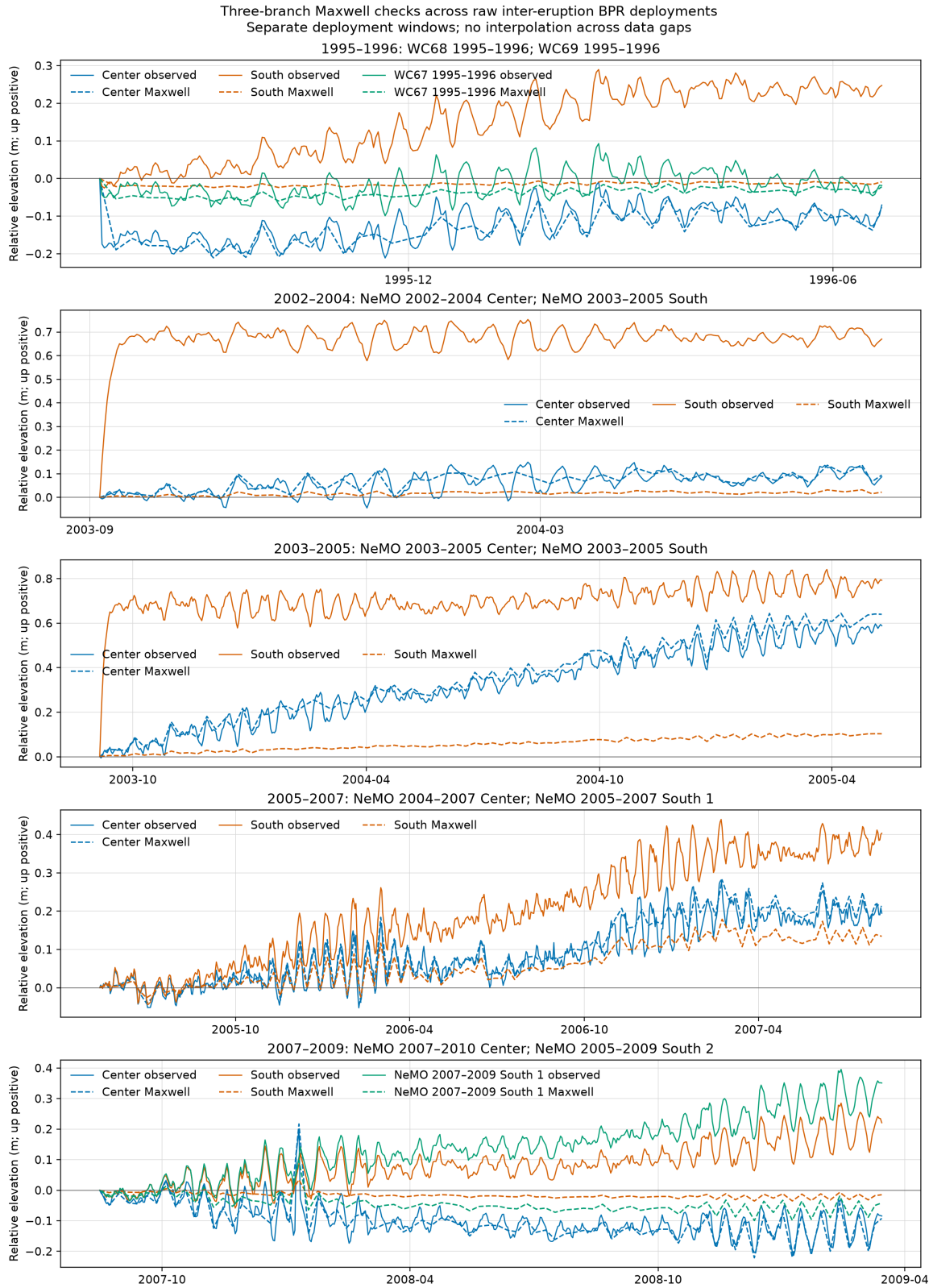


Figure 9: Three-branch Maxwell checks for raw Center/South deployment overlaps from 1995 through 2009. WC67 is held out in 1995–96 and NeMO South 1 in 2007–09. Each pair is independently zeroed; gaps are not interpolated. Generated by `make reproduce`.

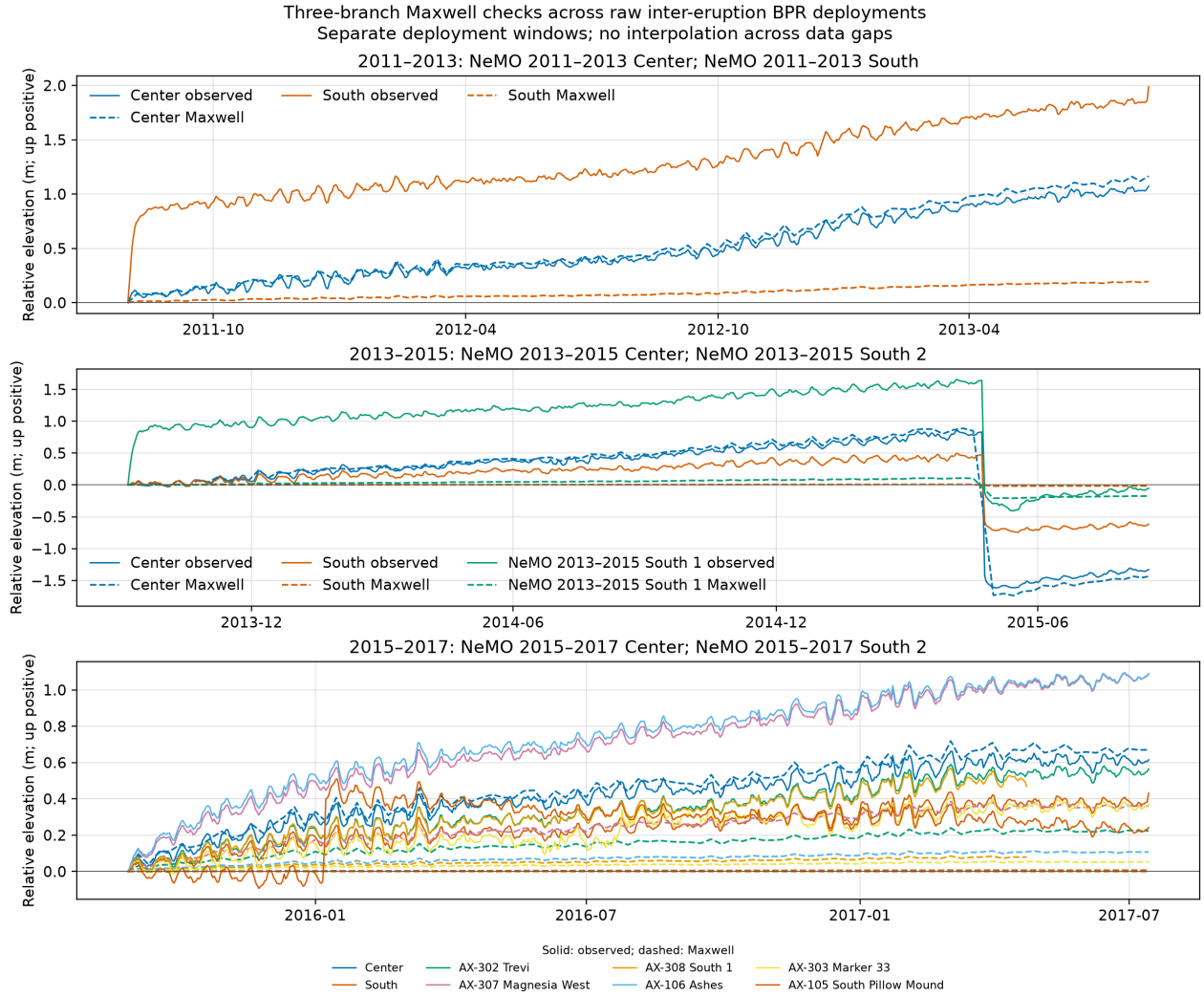


Figure 10: Three-branch Maxwell checks for raw Center/South deployment overlaps from 2011 through 2017. NeMO South 1 is held out for 2013–15. Each pair is independently zeroed; gaps are not interpolated. Center pressure uses static elastic compliance, and synthetic branch parameters and uncorrected records are diagnostic assumptions. The MGDS channels are original **RawDep** data from IEDA/322282, distributed under CC BY-NC-SA 3.0. Generated by `make reproduce`.

| Held-out raw station      | Window  | Days | RMSE (m) | Bias (m) | Correlation |
|---------------------------|---------|------|----------|----------|-------------|
| AX-302 Trevi              | 2015–17 | 687  | 0.211    | −0.189   | 0.984       |
| AX-307 Magnesia West      | 2015–17 | 687  | 0.496    | −0.465   | 0.994       |
| AX-308 South 1            | 2015–17 | 605  | 0.280    | −0.256   | 0.982       |
| AX-106 Ashes              | 2015–17 | 687  | 0.705    | −0.666   | 0.993       |
| AX-303 Marker 33          | 2015–17 | 687  | 0.213    | −0.190   | 0.942       |
| AX-105 South Pillow Mound | 2015–17 | 687  | 0.293    | −0.255   | 0.544       |
| AX-307 Magnesia West      | 2018–20 | 310  | 0.118    | −0.096   | 0.792       |
| AX-302 Trevi              | 2018–20 | 308  | 0.034    | −0.016   | 0.778       |
| AX-105 South Pillow Mound | 2018–20 | 302  | 0.039    | +0.012   | 0.138       |
| BPR West Rim*             | 2018–20 | 739  | 0.346    | +0.305   | −0.267      |
| AX-105 South Pillow Mound | 2020–22 | 648  | 0.180    | +0.174   | 0.174       |
| AX-302 Trevi              | 2020–22 | 648  | 0.031    | −0.014   | 0.722       |
| AX-104 Bag City           | 2020–22 | 647  | 0.047    | +0.031   | 0.554       |
| AX-307 Magnesia West      | 2020–22 | 647  | 0.031    | −0.002   | 0.791       |
| BPR East†                 | 2020–22 | 647  | 0.132    | +0.109   | −0.163      |
| BPR North†                | 2020–22 | 641  | 0.486    | −0.483   | 0.680       |

Table 4: Additional held-out MGDS stations using the existing synthetic three-branch Maxwell histories. All stations remain uncorrected raw observations. \*No MPR-based drift estimate is available for the 2018–20 West record. †Drift is unknown for the 2020–22 full-size BPRs.

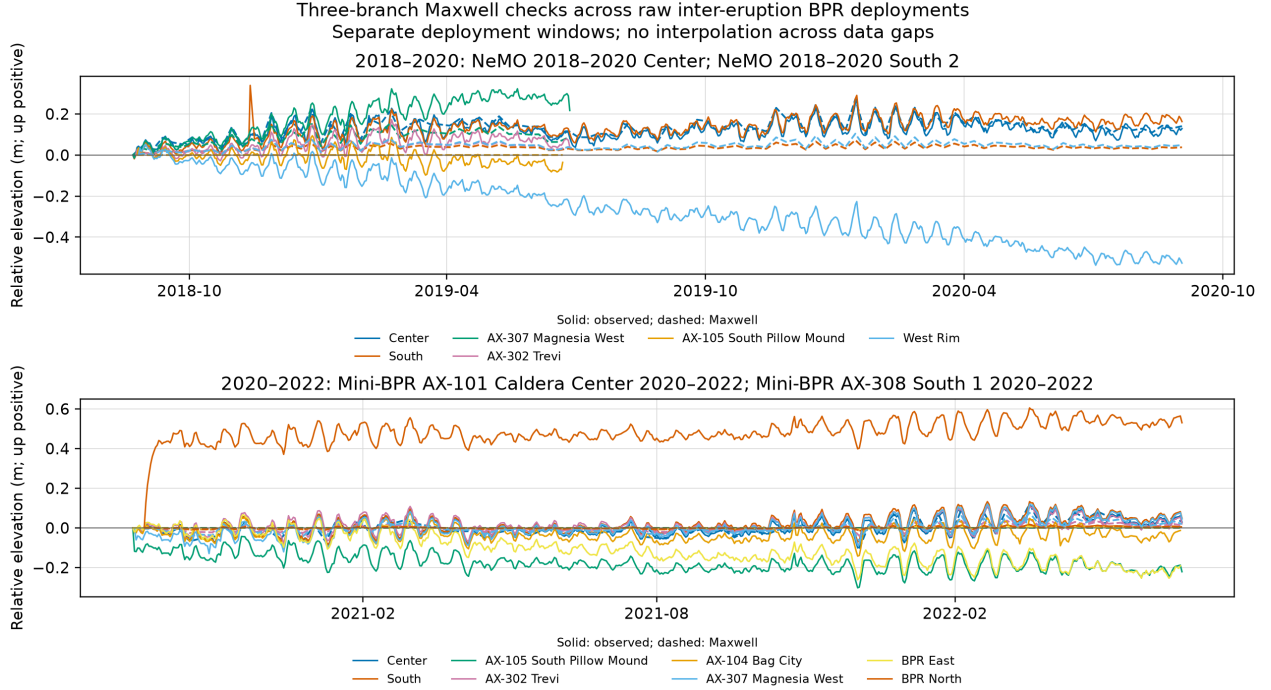


Figure 11: Three-branch Maxwell checks for original raw Center/South deployments from 2018 through 2022, with additional raw station holdouts in both panels. Solid lines are observations and dashed lines are Maxwell predictions. The upper panel includes 15-second moored-BPR `Depth` channels and 100-second miniBPR `RawDep` channels; the lower panel includes 15-second moored-BPR `Depth` and 100-second miniBPR `RawDepth(m)` channels. Each station is independently zeroed, with no pressure-drift or tide correction. Documented location and quality issues are excluded as described in Table 4. Generated by `make_historical-post-2017-bpr-check`.

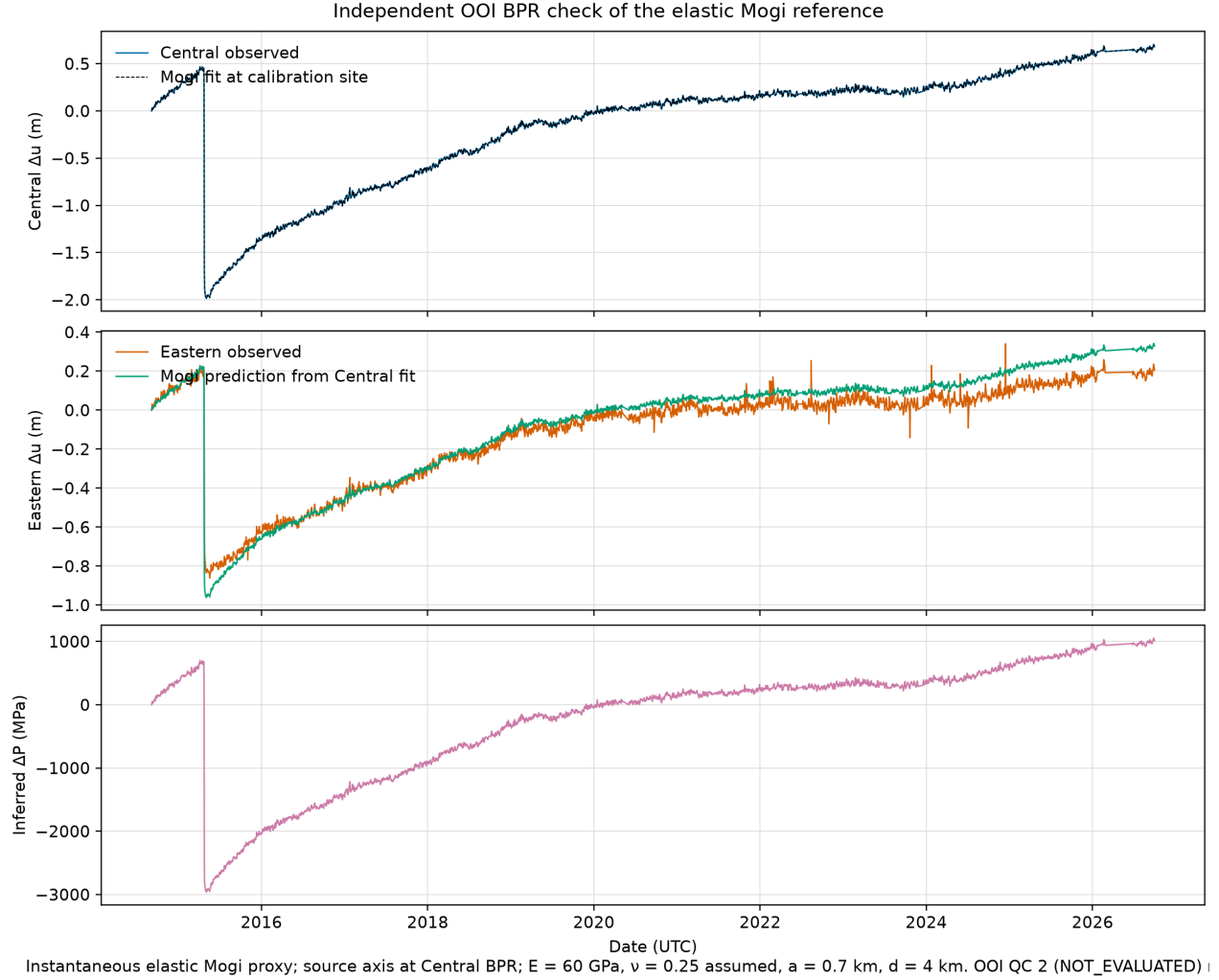


Figure 12: OOI observations and elastic Mogi point-source response. Generated by `make bpr-mogi-check` from independent OOI records; Central is fit by construction and Eastern is held out. The inferred pressure scale is physically unsuitable for the later reservoir model.

The elastic ellipsoid comparison improves the pressure scale but remains mesh-sensitive. On the 2,761-tetrahedron mesh, unit-load compliance is 0.0318994 m/MPa at Central and 0.00345580 m/MPa at Eastern. The static Central-calibrated pressure spans  $-62.21$  to  $22.16$  MPa. Eastern RMSE is 0.221 m, relative L2 error is 76.36 percent, and correlation is 0.995. Across tested refinements, station compliance changes by 21–76 percent; these results do not establish a stable pressure calibration. The bounded station-region comparison uses 2,761, 3,031, 3,124, 3,053, and 3,325 tetrahedra. These independently generated unstructured meshes are not guaranteed to be nested: the 950 m case has fewer elements than the 1,000 m case and changes Central and Eastern compliance by 59.8 and 58.1 percent. The 900 m case then changes them by  $-35.1$  and  $-31.5$  percent. Repeating the five solves reproduced their results exactly, but the sequence has no compliance plateau and no adjacent local-refinement pair meets the 5 percent criterion at both stations.

A separate station-box sequence refines only the Central and Eastern surface sampling neighborhoods and remains below 2,700 tetrahedra. Relative to the 2,761-element baseline, the localized meshes lower compliance by about 63 percent at Central and 55 percent at Eastern. The final 50-to-25 m pair changes compliance by less than 0.1 percent at both sites, but the preceding Eastern changes are 6.55 and 19.85 percent, and element counts are not monotone. This bounded, nonnested sequence does not establish mesh convergence. In the 25 m target mesh, the nearest surface vertices remain 139 m from Central and 102 m from Eastern, so the station sampling locations are not resolved at the target size.

The one-branch Maxwell diagnostic uses monthly pressure inferred from the static elastic Central compliance, linearly interpolated between samples. Its 147 stress records cover 12.07 years. Central RMSE is 1.096 m (correlation 0.787); Eastern RMSE is 0.195 m (correlation 0.922). The Mohr–Coulomb post-processor uses  $C = 1$  MPa, applies the tabulated 25 degrees directly as the friction angle, assumes zero pore pressure, and applies no tensile cutoff because tensile strength is unavailable. It finds a cavity-to-surface path in 146 records, first at a saved time of 60 days. Maximum cavity tension at a saved record with a path is 63.97 MPa. The joint tensile-plus-shear flag remains unset because tensile strength is unknown. These are diagnostic outcomes, not eruption predictions.

Equation 25 writes the friction term as a coefficient, while Table S1 labels the value 25 degrees as an angle. To quantify this ambiguity, I also evaluate literal dimensionless  $f = 25$ , equivalent to a friction angle of 87.71 degrees. Both cases use the same 147 PyLith stress records, cohesion, zero pore pressure, and no tensile cutoff (Table 5). The literal coefficient case already has a connected path at the first saved record, so its onset is not bracketed by an earlier no-path record. Linear stress interpolation places the angle-case onset at 32.02 days, between the 30- and 60-day records. The two outputs show the sensitivity of this provisional postprocessor to the unresolved notation; they do not select an interpretation or predict an eruption.

| Interpretation             | Friction input                     | First path | Path records | Peak cells |
|----------------------------|------------------------------------|------------|--------------|------------|
| Table angle used as $\phi$ | $\phi = 25^\circ$ ( $f = 0.4663$ ) | 60 days    | 146 / 147    | 685        |
| Literal Eq. 25 coefficient | $f = 25$ ( $\phi = 87.71^\circ$ )  | 30 days    | 147 / 147    | 1,345      |

Table 5: Friction-criterion sensitivity in the OOI-driven Maxwell stress history. Path counts use the same stress fields and mesh; the shear path does not apply a tensile cutoff.

The static pressure conversion does not account for Maxwell memory. A second OOI-only check instead fits Central uplift directly through a PyLith one-branch Maxwell response kernel and reserves Eastern for a spatial holdout. For this assumed rheology, the Central RMSE is 0.00495 m and Eastern RMSE is 0.225 m (correlation 0.983). The inferred pressure spans  $-60.4$  to  $11.8$  MPa. The direct PyLith history agrees with kernel superposition to relative L2 errors of 0.00028 at Central and 0.00080 at Eastern. This verifies the linear response implementation; it does not validate the pressure scale, which still depends

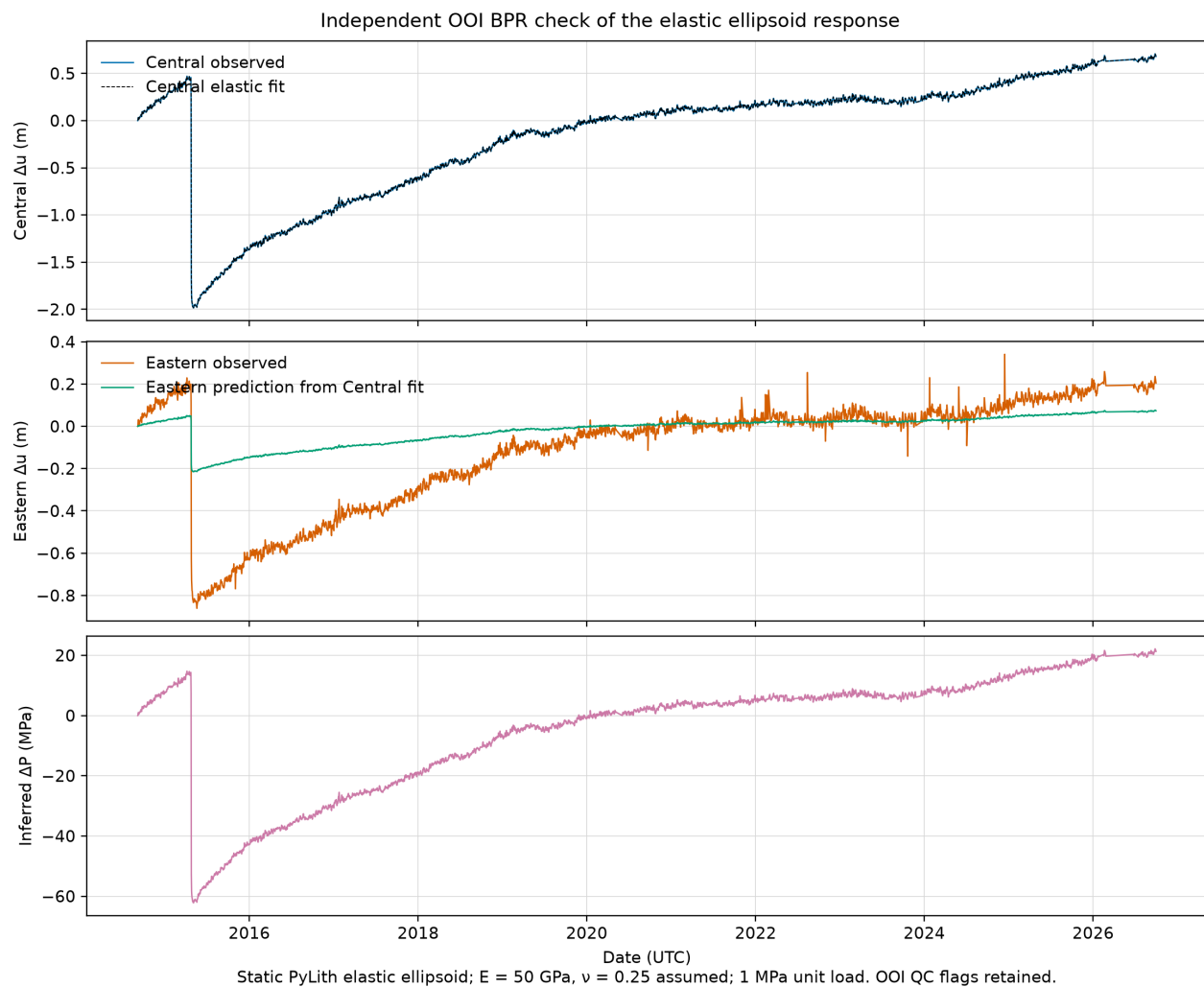
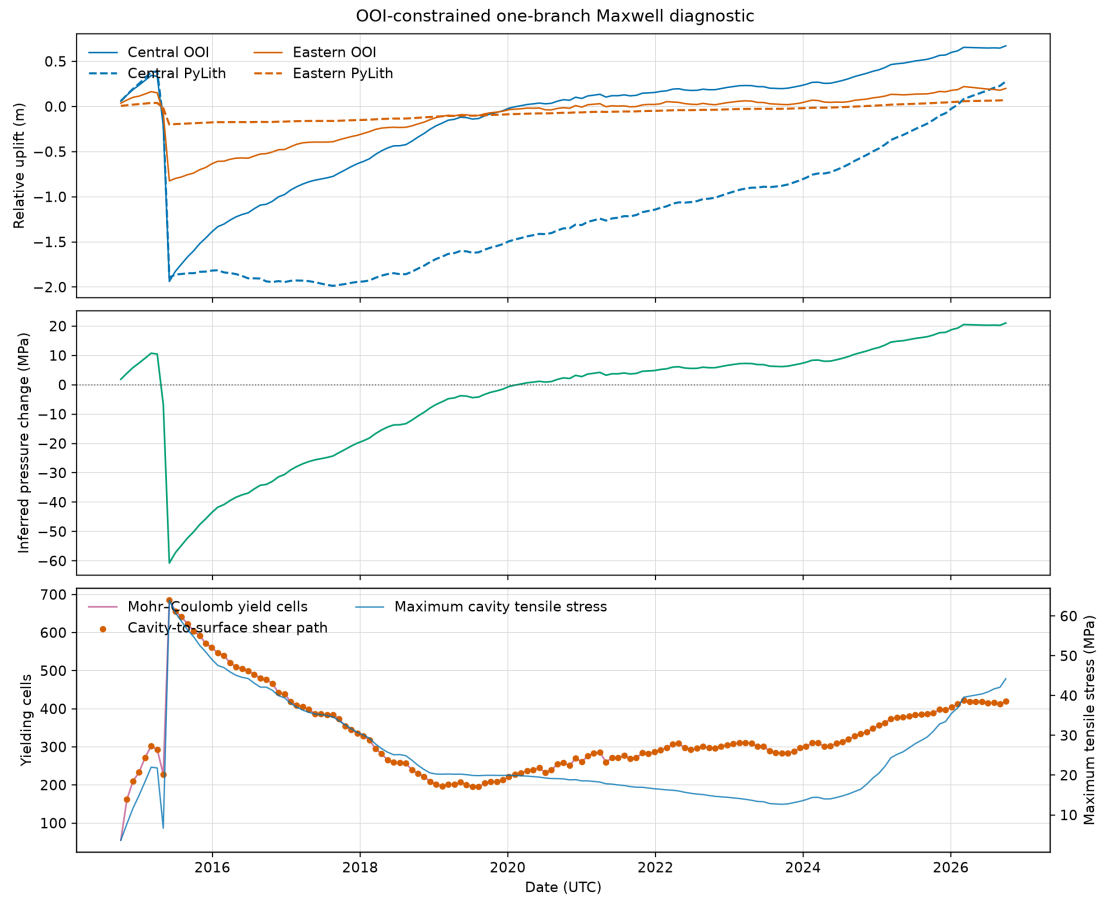


Figure 13: Static elastic ellipsoid fit to Central OOI uplift and Eastern holdout prediction. Generated by `make ellipsoid-bpr-check`. The amplitude mismatch and nonconverged compliance make the inferred pressure provisional.



OOI QC 2 (NOT\_EVALUATED) retained. Pressure uses static elastic Central compliance. Failure uses  $C = 1$  MPa,  $25^\circ$  directly, zero pore pressure, and no tensile cutoff. The 2,761-tetrahedron compliance is not mesh-converged.

Figure 14: OOI-constrained one-branch Maxwell diagnostic. The panels compare OOI and modeled uplift, static-fit pressure, and stress-derived failure indicators. Generated by `make ooi-maxwell-history-plot`. Quality code 2 is retained; pressure uses static compliance; the mesh and failure assumptions remain provisional.

on a nonconverged mesh, GCV smoothing, and an assumed single Maxwell branch.

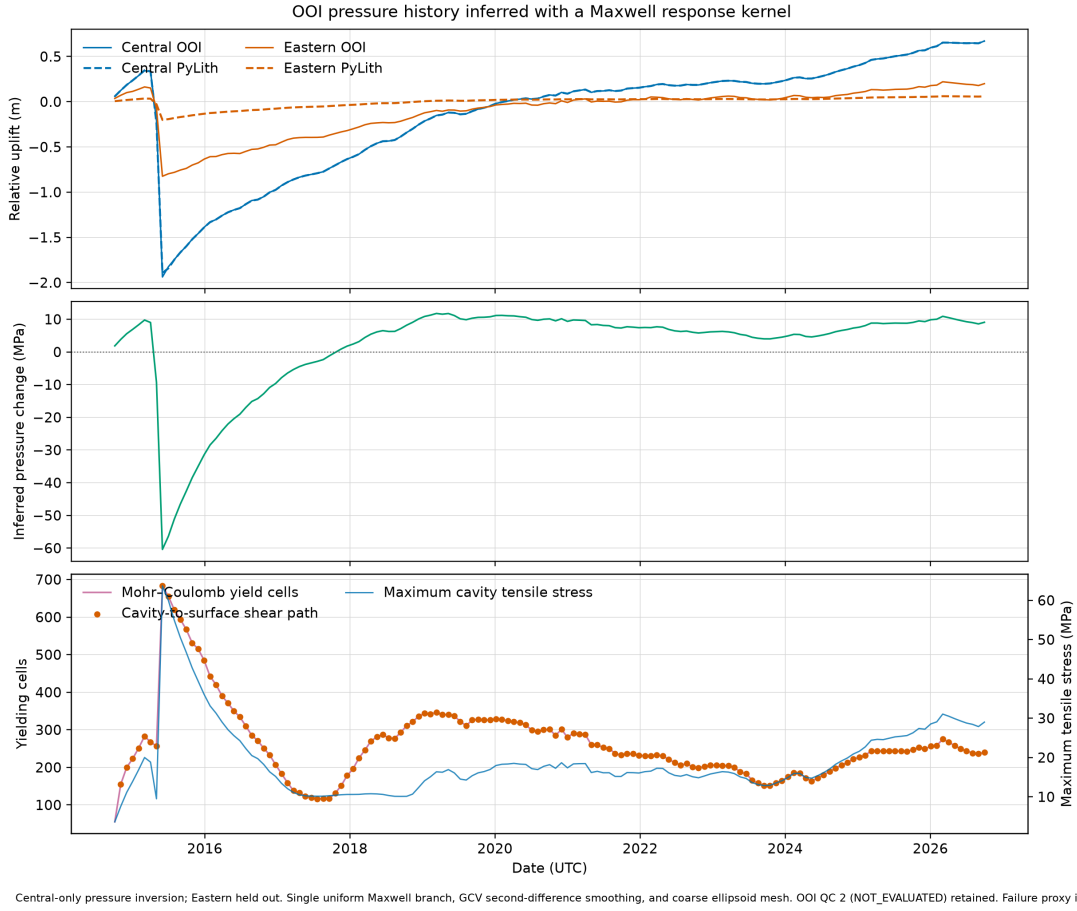


Figure 15: OOI Central pressure inversion with a one-branch Maxwell response kernel. Central is fitted and Eastern is held out; dashed curves show the direct PyLith history. The plot includes the inferred pressure and provisional failure diagnostics. Generated by `make ooi-maxwell-pressure-inversion` from independent OOI records. The smoothness prior, pressure scale, and nonconverged mesh remain provisional.

The response-kernel check also extends to the earlier eruption records using original non-OOI BPR channels. WC81/WC82A span 309 paired days across 1998; NeMO Center/South span 314 days across 2011. Central RMSE is 0.093 m for 1998 and 0.125 m for 2011, while the South holdout RMSE is 0.535 and 0.713 m with biases of +0.362 and +0.385 m. The fitted pressure minima are  $-107.4$  and  $-71.6$  MPa. Direct PyLith histories match the unit-ramp kernels to below 0.13 percent relative L2 error. The Southern residuals and large pressure amplitudes show that the one-branch rheology and coarse mesh do not calibrate a physical eruption pressure history.

The four-case pressure-calibrated comparison fits raw Center records independently under static elasticity, constant-property generalized Maxwell, baseline temperature-dependent Maxwell, and hydrothermal temperature-dependent Maxwell rheologies. For 1998, it uses raw NCEI absolute-pressure channels from WC81 Center and WC82A South, with 309 paired days from 3 October 1997 through 7 August 1998. Center RMSE is 0.093 m; held-out South RMSE is 0.519–0.530 m with +0.350 to +0.359 m bias. All four cases show a connected path at the first saved 7-day record, which bounds proxy onset but does not resolve timing relative to the eruption. For 2011, the raw MGDS NeMO pair yields 0.124 m Center RMSE and 0.704–0.717 m South RMSE with approximately +0.38 m bias. Direct Maxwell histories

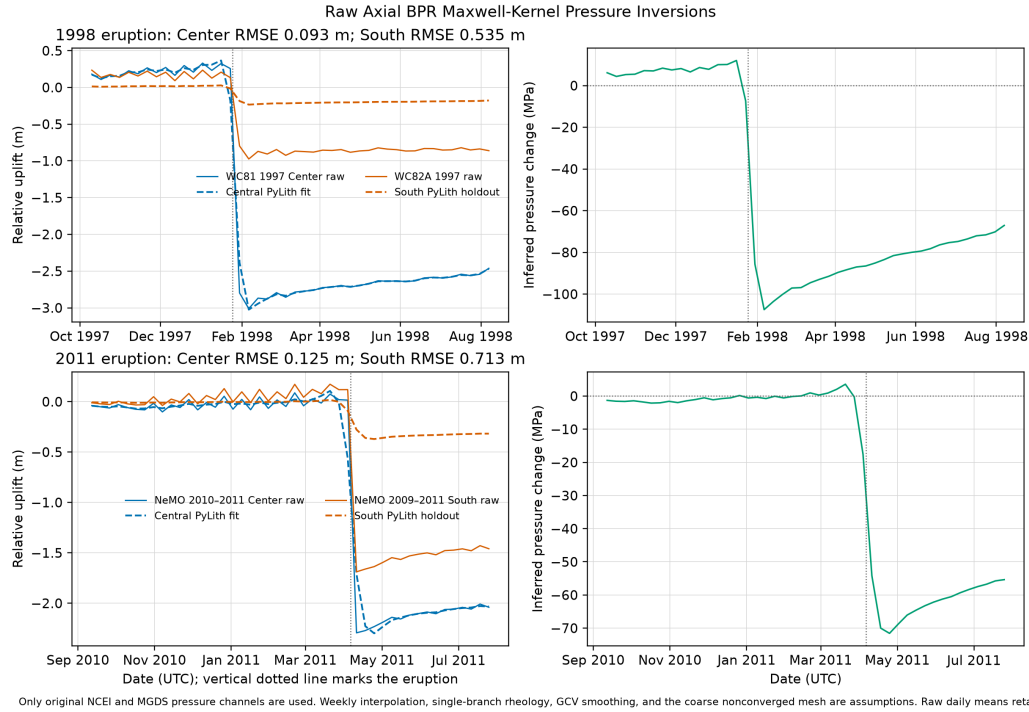


Figure 16: Maxwell-kernel pressure inversions of original raw BPR data across the 1998 and 2011 eruptions. Solid curves are uncorrected daily observations; dashed curves show the Central fit and held-out South prediction. Vertical dotted lines mark the eruption dates. Generated by `make_historical-bpr-maxwell-pressure-inversion`. Sources are NOAA/NCEI DART raw channels (doi:10.7289/V5F18WNS) and original MGDS IEDA/322282 `Depth/RawDep` channels (William Chadwick, Scott Nooner, and MGDS; CC BY-NC-SA 3.0).

reproduce ramp-kernel predictions to below 0.22 percent relative L2 error in both windows. The 2011 pressure minima range from  $-72$  MPa for elasticity and  $-67$  MPa for constant-property Maxwell to about  $-34$  MPa for the temperature-dependent cases. Its provisional connected paths first appear about 35, 38, and 205 days after the 5 September 2010 record start. Both Center fits include eruption deflation and later records, so these are retrospective diagnostics. Synthetic Maxwell branches, Eq. 16 as printed, the assumed thermal boundaries, and nonconverged compliance limit these results. The method and complete diagnostics are documented in `pylith/step15_historical_four_case_bpr/README.md`.

## 4 Numerical verification

The bounded PyLith 5.0.2 toolchain runs an elastic-cavity smoke case on 2,761 linear tetrahedra. Analytical Mogi checks verify positive inflation uplift, radial symmetry, and pressure scaling. A PyLith spherical-cavity comparison has a 33.2 percent nearest-axis error on 3,191 tetrahedra; this coarse check does not establish mesh convergence. Increasing the half-width and bottom depth from 8 km to 12 km reduces peak uplift by 61.4 percent and raises fixed-grid vector error from 40.4 to 58.0 percent on independently generated, nonnested meshes. This comparison does not isolate the finite-boundary effect; neither mesh validates the analytical response quantitatively. The tetrahedral thermal solver matches manufactured linear and uniform-source solutions. Its full ellipsoid thermal case converges numerically, but its lateral and basal boundary temperatures remain assumptions.

The failure analysis converts PyLith Cauchy stress to principal stresses, applies a Mohr–Coulomb yield test, and searches face-connected tetrahedra between the cavity and surface. Synthetic tests verify state ordering, boundary identification, and time-resolved path detection. Those checks validate software behavior under stated assumptions; they do not validate a physical failure threshold.

## 5 Panel-by-panel status

No manuscript numerical panel is independently reproduced. Figure 4 is an observation-only partial record; Figures 12–16 are project-generated diagnostics. The detailed command, output, and comparison inventory is maintained in `docs/figure_reproduction.md`.

| Panel                         | Status and current limitation                                                                                                                                                                                                                                                                                                              |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fig. 1a                       | Not reproduced; required bathymetry, lava-flow, and event source records are excluded.                                                                                                                                                                                                                                                     |
| Fig. 1b                       | Schematic only; the fallback box and unresolved boundary conventions preclude a numerical panel comparison.                                                                                                                                                                                                                                |
| Fig. 2                        | Partially reproduced from OOI records for 2014–2026 and intermittent raw BPR deployments for 1987–2022; generalized Maxwell checks span twelve windows, including the 1998/2011 events and ten inter-eruption intervals. NCEI adds spatial checks across 1987–1996. The continuous history, eruption markers, and event counts are absent. |
| Fig. 3, elastic               | Not reproduced; the required property slices and full written configuration are unavailable.                                                                                                                                                                                                                                               |
| Fig. 3, viscoelastic          | Not reproduced; a one-branch smoke test does not provide generalized rheology or panel fields.                                                                                                                                                                                                                                             |
| Fig. 3, temperature-dependent | Not reproduced; Eq. 16 is inconsistent with its stated limits and the fields are incomplete.                                                                                                                                                                                                                                               |
| Fig. 3, hydrothermal          | Partially reproduced; one hydrothermal property slice is available, but the full rheology panels and resolved Eq. 16 are not.                                                                                                                                                                                                              |

| Panel     | Status and current limitation                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fig. 4a   | Not reproduced; static, one-branch, and three-branch checks cover raw deployments from 1987–2022 and OOI from 2014–2026, including 1998 and 2011. Four-case raw fits use WC81/WC82A in 1998 and NeMO Center/South in 2011; Center RMSE is 0.093/0.124 m and held-out South RMSE is 0.519–0.531/0.704–0.717 m. Both fits include eruption deflation; a calibrated full-cycle history and mesh-converged compliance remain unavailable. |
| Fig. 4b   | Not reproduced; the full calibrated reservoir-volume and historical deformation record is unavailable.                                                                                                                                                                                                                                                                                                                                |
| Fig. 5a   | Not reproduced; event-scale stress history and rheology mapping are unavailable.                                                                                                                                                                                                                                                                                                                                                      |
| Fig. 5b   | Not reproduced; event-scale stress history and rheology mapping are unavailable.                                                                                                                                                                                                                                                                                                                                                      |
| Fig. 5c   | Not reproduced; event-scale stress history and rheology mapping are unavailable.                                                                                                                                                                                                                                                                                                                                                      |
| Fig. 5d   | Not reproduced; event-scale stress history and rheology mapping are unavailable.                                                                                                                                                                                                                                                                                                                                                      |
| Fig. 5e   | Not reproduced; event-scale stress history and rheology mapping are unavailable.                                                                                                                                                                                                                                                                                                                                                      |
| Fig. 5f   | Not reproduced; event-scale stress history and rheology mapping are unavailable.                                                                                                                                                                                                                                                                                                                                                      |
| Fig. 5g   | Not reproduced; event-scale stress history and rheology mapping are unavailable.                                                                                                                                                                                                                                                                                                                                                      |
| Fig. 5h   | Not reproduced; event-scale stress history and rheology mapping are unavailable.                                                                                                                                                                                                                                                                                                                                                      |
| Fig. 5i   | Not reproduced; event-scale stress history and rheology mapping are unavailable.                                                                                                                                                                                                                                                                                                                                                      |
| Fig. 5j   | Not reproduced; event-scale stress history and rheology mapping are unavailable.                                                                                                                                                                                                                                                                                                                                                      |
| Fig. 5k   | Not reproduced; event-scale stress history and rheology mapping are unavailable.                                                                                                                                                                                                                                                                                                                                                      |
| Fig. 5l   | Not reproduced; event-scale stress history and rheology mapping are unavailable.                                                                                                                                                                                                                                                                                                                                                      |
| Fig. 5m   | Not reproduced; event-scale stress history and rheology mapping are unavailable.                                                                                                                                                                                                                                                                                                                                                      |
| Fig. 5n   | Not reproduced; event-scale stress history and rheology mapping are unavailable.                                                                                                                                                                                                                                                                                                                                                      |
| Fig. 5o   | Not reproduced; event-scale stress history and rheology mapping are unavailable.                                                                                                                                                                                                                                                                                                                                                      |
| Fig. 5p   | Not reproduced; event-scale stress history and rheology mapping are unavailable.                                                                                                                                                                                                                                                                                                                                                      |
| Fig. 5q   | Not reproduced; event-scale stress history and rheology mapping are unavailable.                                                                                                                                                                                                                                                                                                                                                      |
| Fig. 5r   | Not reproduced; event-scale stress history and rheology mapping are unavailable.                                                                                                                                                                                                                                                                                                                                                      |
| Fig. 5s   | Not reproduced; event-scale stress history and rheology mapping are unavailable.                                                                                                                                                                                                                                                                                                                                                      |
| Fig. 5t   | Not reproduced; event-scale stress history and rheology mapping are unavailable.                                                                                                                                                                                                                                                                                                                                                      |
| Fig. 6a–d | Not reproduced; no conceptual redraw has been made.                                                                                                                                                                                                                                                                                                                                                                                   |
| Supp. S1  | Partially reproduced for one hydrothermal temperature-dependent case; complete slices and all four rheologies are not available.                                                                                                                                                                                                                                                                                                      |
| Supp. S2  | Not reproduced; the hydrothermal brittle–ductile transition comparison is incomplete.                                                                                                                                                                                                                                                                                                                                                 |
| Supp. S3  | Not reproduced; independent Del Negro and prior FEM benchmarks are incomplete.                                                                                                                                                                                                                                                                                                                                                        |
| Supp. S4  | Not reproduced; fixed-base depth sensitivity is nonmonotonic on nonnested meshes and does not establish Winkler equivalence.                                                                                                                                                                                                                                                                                                          |
| Supp. S5  | Schematic only; side and basal geotherm, fallback domain, and tectonic face-rate split remain qualified assumptions.                                                                                                                                                                                                                                                                                                                  |
| Supp. S6  | Not reproduced; mesh-converged OOI pressure calibration and the full rheology set are unavailable.                                                                                                                                                                                                                                                                                                                                    |

## 6 Limitations and next work

The present implementation is a collection of verified components, OOI checks, and raw historical BPR diagnostics. It runs the four rheology configurations on one pressure-calibrated 2011 window, but it does not recover a generalized Maxwell spectrum, a verified Winkler foundation, or the complete historical pressure history. The model-box dimensions, Poisson ratio, host density, tensile strength, and several source conventions remain unresolved. The finite-element compliance is not mesh-converged.

A fixed-base depth sensitivity check compares 20, 30, and 40 km domains while embedding both BPR coordinates as surface vertices. The meshes contain 2,601, 2,774, and 2,869 tetrahedra. Relative to 20 km, Central compliance changes by +6.54% at 30 km and –1.34% at 40 km; Eastern compliance

1998 raw BPR pressure calibration across four written rheologies  
synthetic Maxwell branches; Eq. 16 used as printed; South held out

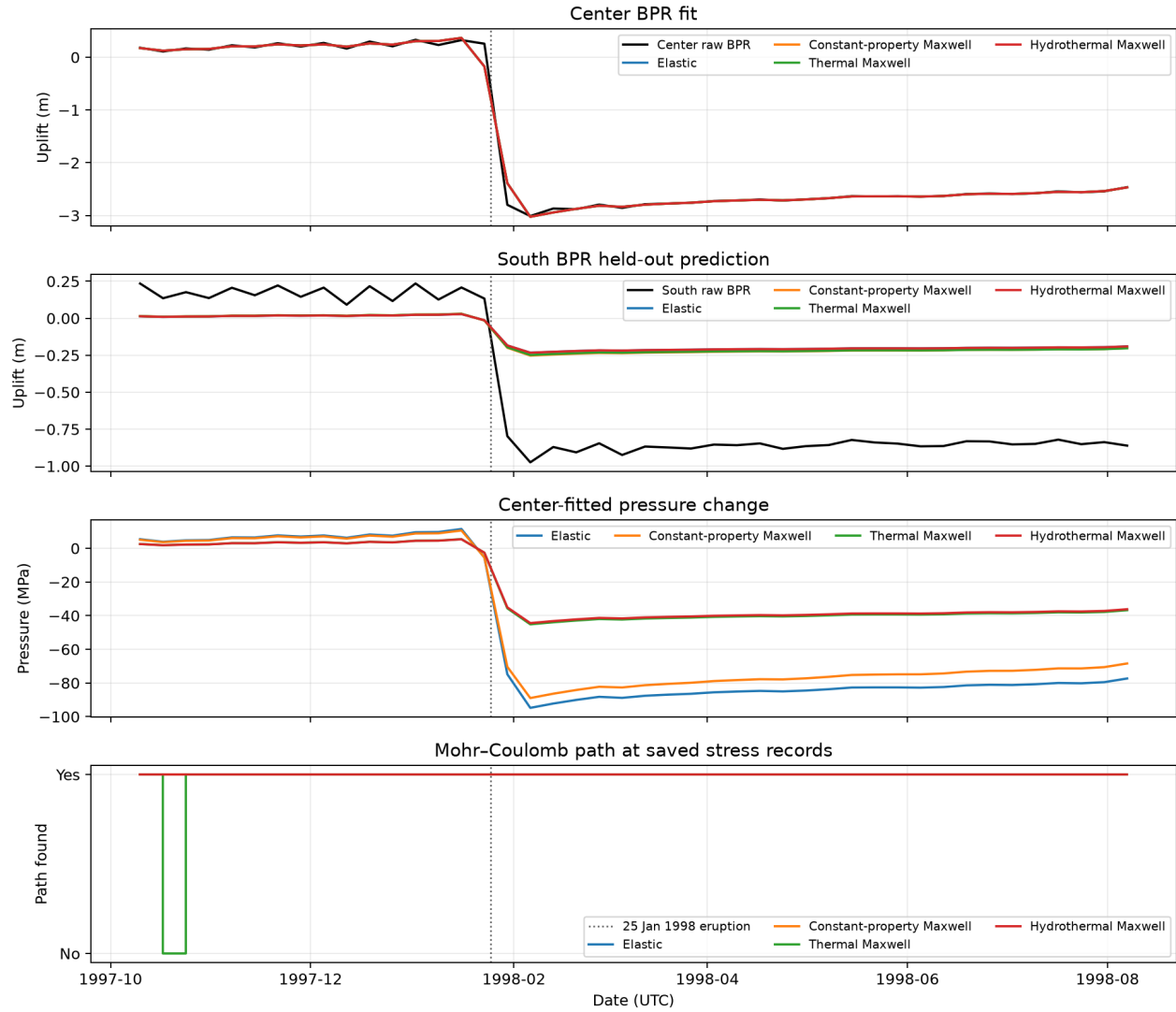


Figure 17: Four pressure calibrations against original raw NCEI WC81 Center data, with WC82A South held out and provisional connected-path states shown at saved stress records. The dotted line marks the observed 25 January 1998 eruption. Each pressure history is fitted independently, and Center includes the eruption deflation and later records. Synthetic branch parameters, the nonconverged mesh, and unresolved failure assumptions make this a retrospective diagnostic. No MGDS Fox duplicate or Cabaniss-associated data products were used. Generated by `make historical-four-case-bpr-calibration`.

2011 raw BPR pressure calibration across four written rheologies  
synthetic Maxwell branches; Eq. 16 used as printed; South held out

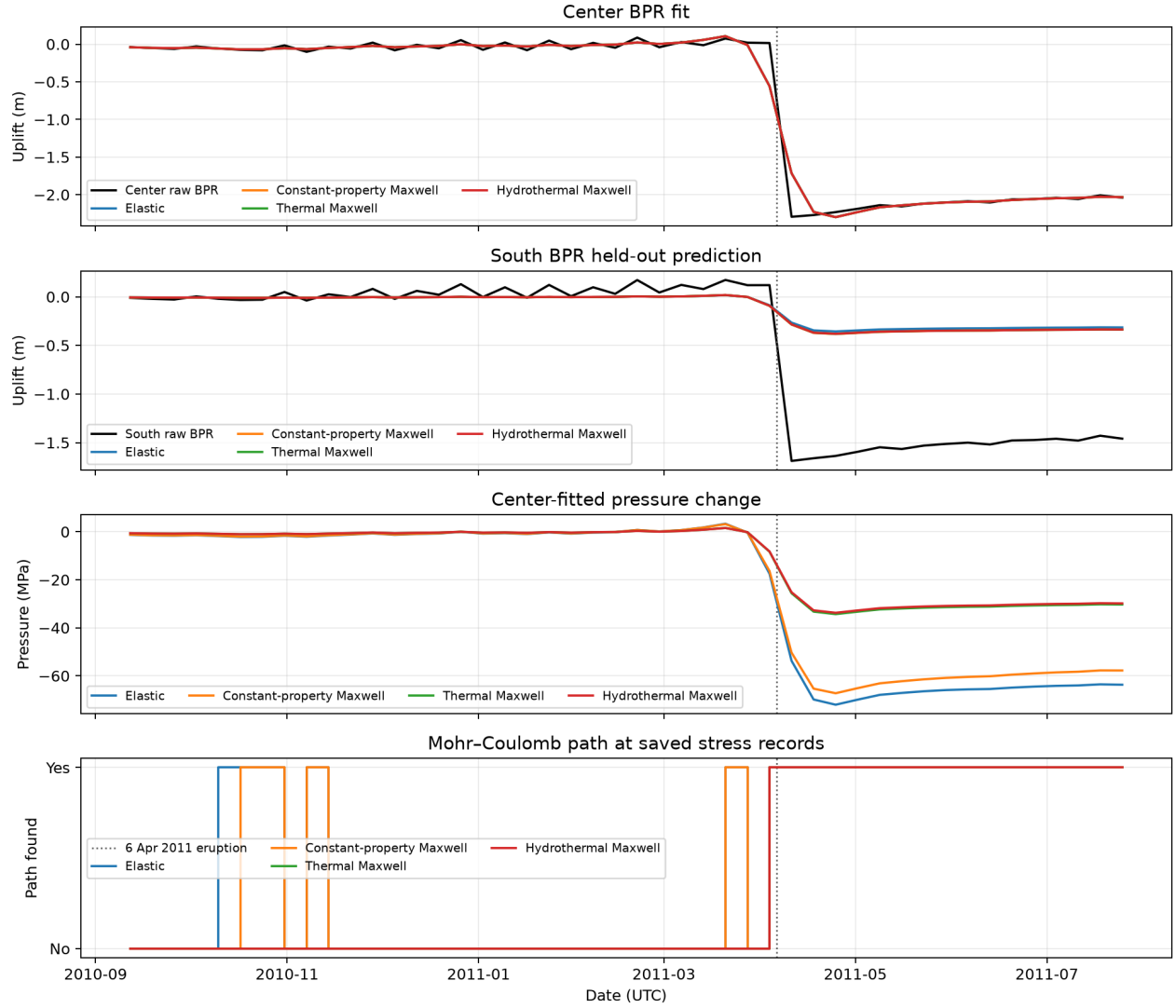


Figure 18: Four pressure calibrations against original raw NeMO Center BPR data, with NeMO South held out and provisional connected-path states shown at saved stress records. Dotted vertical lines mark the observed 6 April 2011 eruption. Each pressure history is fitted independently; the Center input includes the eruption deflation and subsequent data. Synthetic branch parameters, the nonconverged mesh, and unresolved failure assumptions make this a retrospective diagnostic. Generated by `make historical-four-case-bpr-calibration`. Sources are original MGDS IEDA/322282 RawDep/Depth channels (William Chadwick, Scott Nooner, and MGDS; CC BY-NC-SA 3.0).

changes by  $-7.30\%$  and  $+4.66\%$ , respectively. These nonmonotonic responses on nonnested meshes do not demonstrate domain convergence or Winkler equivalence.

Original raw NCEI channels from ten deployments extend the static check to 1987–1996. A 43-day WC51-fit/WC61-holdout comparison has 0.020 m RMSE and 0.685 correlation. The 338-day WC68-fit checks against WC69 and WC67 have 0.183 m and 0.038 m RMSE. Single-station fits reach  $+294$  MPa; drift, ocean variability, separate instrument baselines, and nonconverged compliance remain unresolved. These raw observations expand temporal coverage but do not provide a physical pressure estimate or a continuous pre-1998 history.

The written thermal equation is steady and sets heat production to zero, while the project roadmap asks for mechanics-to-thermal feedback. No return-coupling law is specified in the written model. This report preserves that gap instead of silently inventing a source term. A scientifically faithful reproduction cannot claim full two-way coupling until the governing feedback and missing parameters are specified. Raw historical BPR checks add eruption and inter-eruption observations, but their uncorrected channels and static elasticity do not reproduce the 1998–2011 stress evolution.

Static PyLith ellipsoid checks across five raw overlap windows from 1995–2013 yield seven held-out station RMSE values: 0.180/0.028 m (WC69/WC67, 1995–96), 0.257 m (2003–05), 0.166 m (2005–07), 0.124/0.244 m (South 2/South 1, 2007–09), and 0.614 m (2011–13). The 2011–13 comparison has  $+0.551$  m bias despite 0.993 correlation; NeMO South 1 is anticorrelated at  $-0.325$ . Raw ocean and instrument variability and the unconverged mesh limit interpretation.

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